

通信与网络课程设计

设计开发DNS Relay

感谢北邮计算机学院高占春副教授
提供原版课件



主讲：刘洋 副教授

北京邮电大学 电子工程学院

Email: ly@bupt.edu.cn

■ 课程介绍

- 本学期通信与网路课程设计为1.5学分，教学内容为综合课实验，教学方式采用开放式实验教学模式，第19周上课。
- 以目前广泛使用的TCP/IP协议栈为基础，使用UDP或TCP通信平台，利用Socket编程界面，实现DNS中继器软件或TFTP文件传送协议（Trivial File Transfer Protocol）服务器端软件。通过本课程的学习，使学生熟练掌握Socket编程界面的使用、能够对通信协议进行描述和设计、掌握计算机网络协议设计、编程和测试的基本方法和相关技术，提高计算机网络与通信系统的设计与开发和实际环境中设计开发系统的能力。

- 基本要求
- DNS是什么
- 如何使用Wireshark软件
- 如何进行Client-Server Socket通信软件开发

- 从DNS Client接收到DNS查询请求并转发DNS查询请求到一个给定的DNS Server
- 从DNS Server接收到DNS查询响应并转发给DNS解析器 (DNS Resolver)



- 查找本地数据库（例如dnsrelay.txt）
 - Case 1：对于IP地址0.0.0.0, 发送返回“no such name”
(reply code = 0011)
 - Case 2：对于包括在本地数据库里的domain name, 发送返回相应的IP地址
 - Case 3：对于没有包括在本地数据库里的domain name, 转发查询请求给本地的DNS服务器 (DNS Server)

- 操作系统 (Operating Systems)
 - Windows, Linux
- 编程语言
 - C/C++, Python, Java,
- 工作小组
 - 每个小组三个人
 - 一个源程序和一份实验报告

■ Domain Name System

■ 分布式数据库文件提供域名（Domain Name）和IP地址之间的联结映射

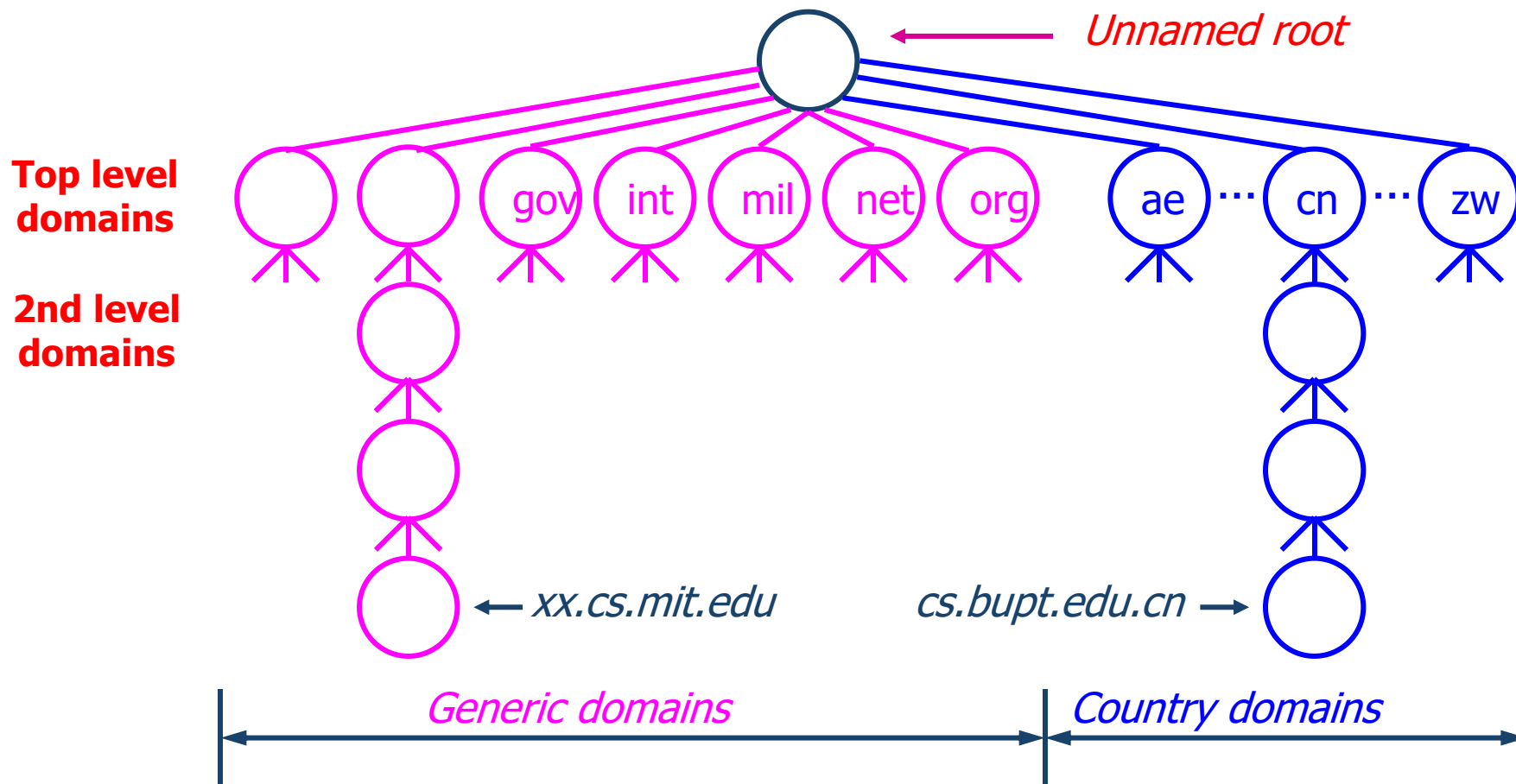
- Implemented in hierarchy of many name servers

■ 应用层协议

- 由hosts和Name servers通信交互来解析域名（Domain Name）

■ 多在传输层（transport layer）使用UDP协议

Domain Namespace-分层结构

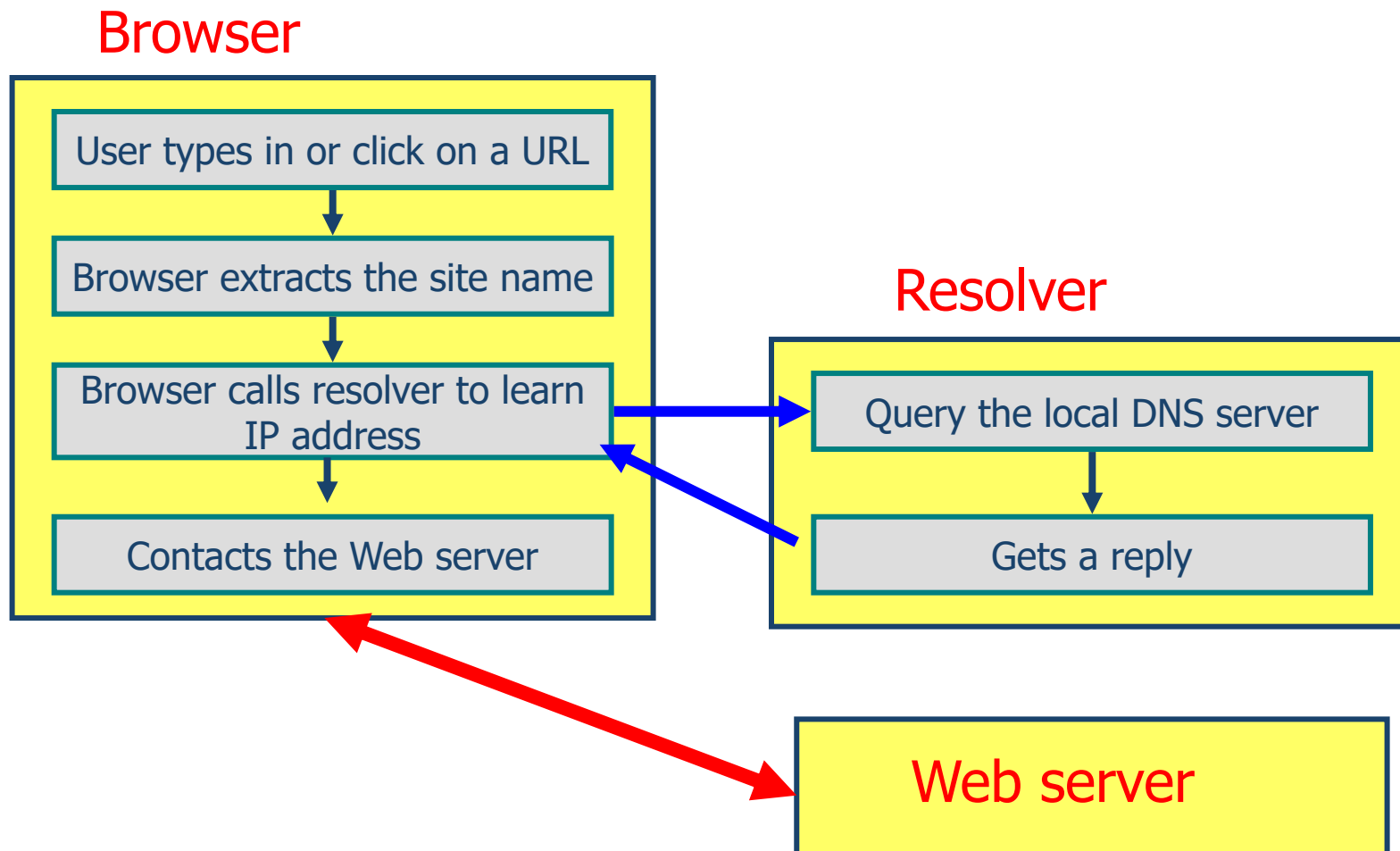


- Name Servers是组成域名数据库 (Domain Database) 的信息容器
- 回答关于域名信息的查询请求

■ 是用户程序和域名服务器之间的接口

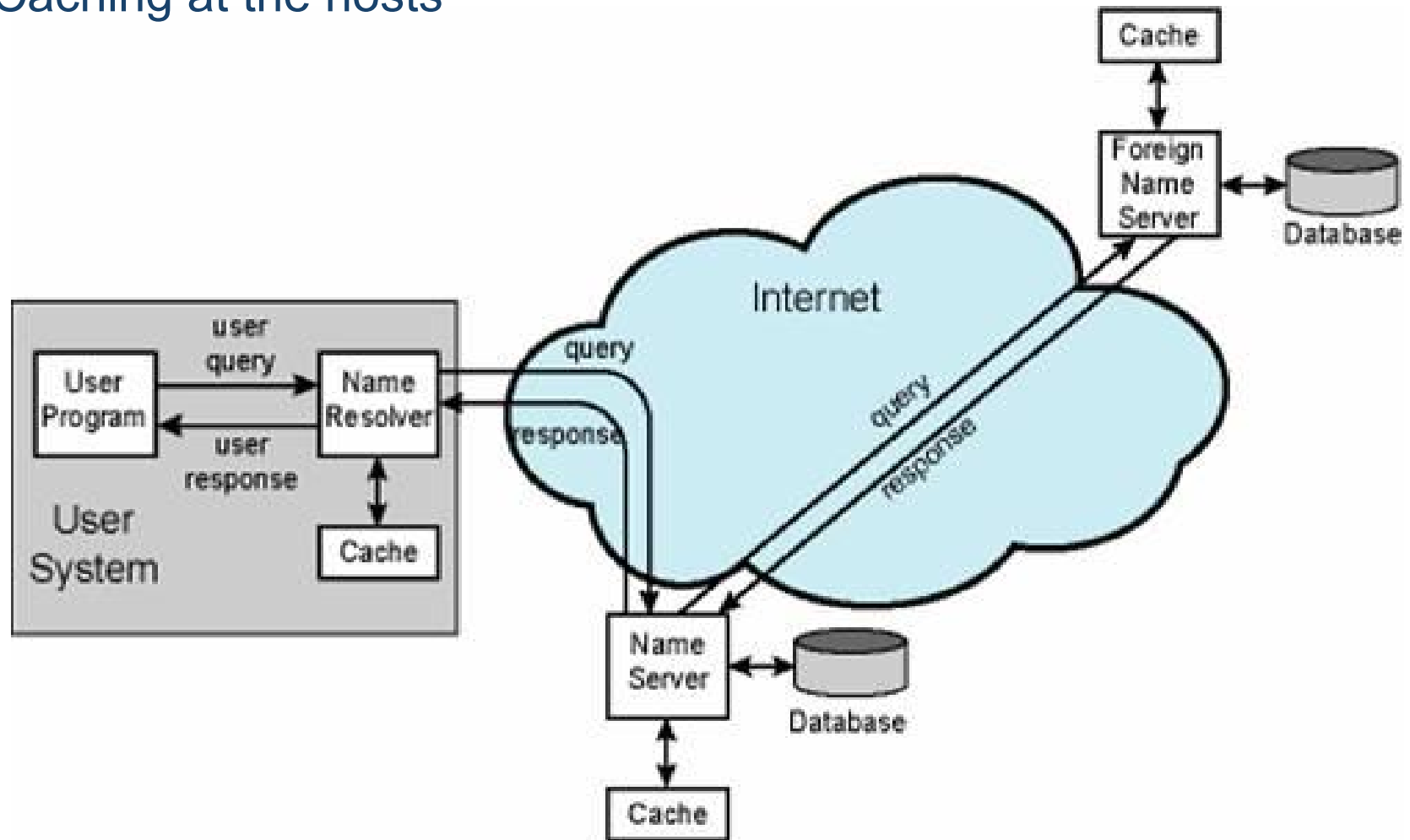
- resolver接收来自用户程序的请求 (例如, http、mail programs、telnet、ftp、ping)
- Asks questions to the DNS system on behalf of the application
- 返回想要的信息

举例：DNS和HTTP application交互



- ❖ **DNS**中的每一个域（**Domain**）由一个或多个**Resource Records (RRs)**, 包含了域名中的字段信息
- ❖ 每个**Resource Records**包含以下信息:
 - **Owner**: the domain name where the RR is found
 - **Type**: specifies the type of the resource in this RR
 - A – Host Address
 - MX – Mail Exchanger
 - ...
 - **Class**: specifies the protocol family to use
 - IN – the Internet system
 - **TTL**: specifies the Time To Live (in unit of second) of the cached RRs
 - **RDATA**: the resource data

- Caching at the name servers
- Caching at the hosts



协议细节：消息格式（根据RFC 1035）

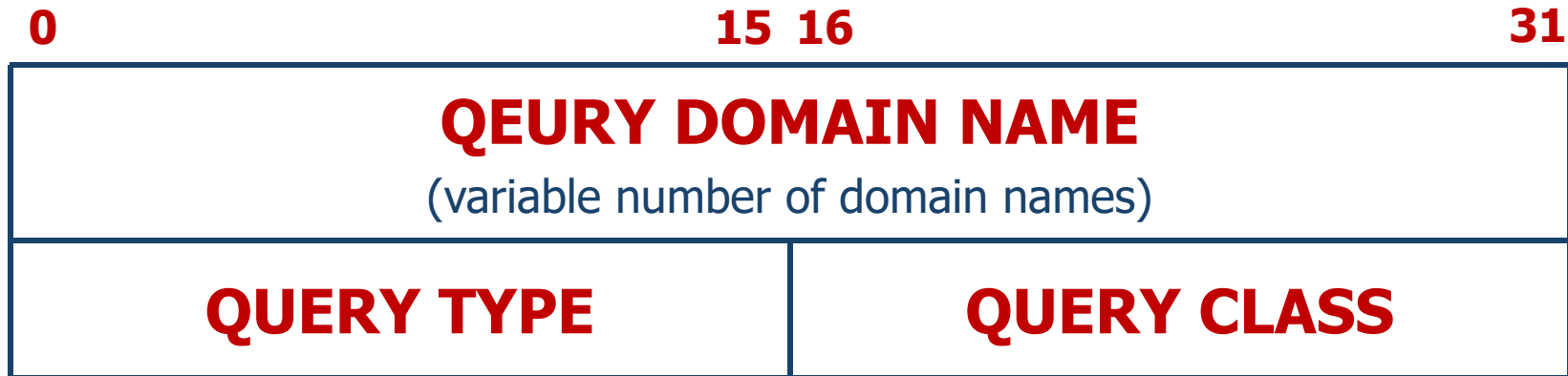


	0	15	16	Flags					31
Fixed length	ID	QR	OPCODE	AA	TC	RD	RA	Z	Rcode
	Question count	Answer count							
	Authority count	Additional count							
Variable length	Question Section (variable number of questions) ➡								
	Answer Section (variable number of Resource Recordss) ➡								
	Authority Section (variable number of Resource Recordss)								
	Additional Section (variable number of Resource Recordss)								

14

- **ID**: 16-bit field used to correlate queries and responses.
- **QR**: 1-bit field that identifies the message as a query (0) or response (1).
- **OPCODE**: 4-bit field that describes the type of query:
 - 0: Standard query (name to address). 1: Inverse query (address to name). 2: Server status request.
- **AA**: Authoritative Answer. 1-bit field. When set to 1, identifies this response is made by an authoritative name server.
- **TC**: Truncation. 1-bit field. When set to 1, indicates the message has been truncated due to length greater than that permitted.
- **RD**: Recursion Desired. 1-bit field. Set to 1 by the resolver to request recursive service by the name server.
- **RA**: Recursion Available. 1-bit field. Set to 1 by name server to indicate recursive query support is available.
- **Z**: 3-bit field. Reserved for future use. Must be set to 0.

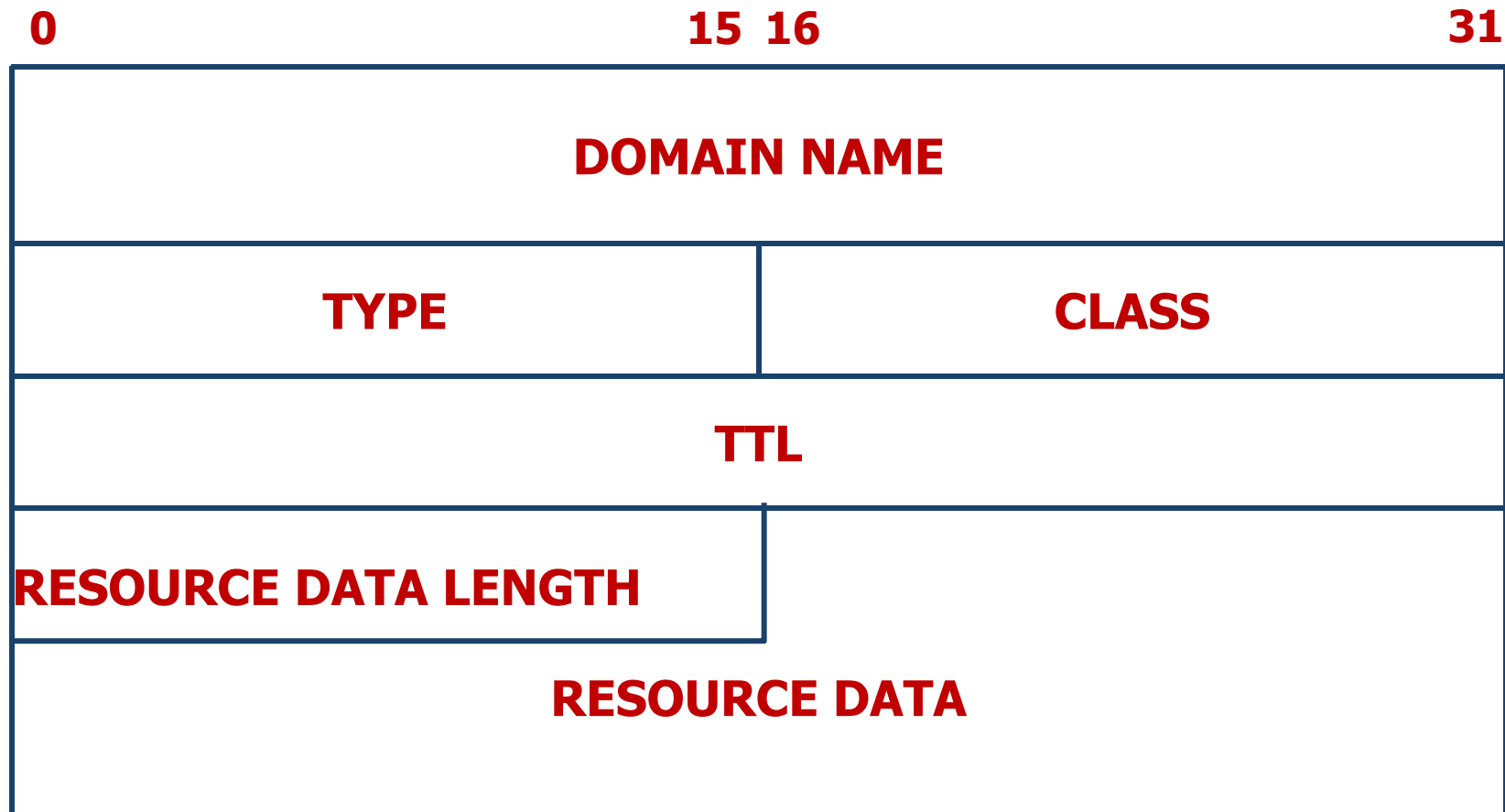
- **RCODE**: Response Code. 4-bit field that is set by the name server to identify the status of the query:
 - 0: No error condition. 1: Unable to interpret query due to format error. 2: Unable to process due to server failure. 3: **Name in query does not exist.** 4: Type of query not supported. 5: Query refused for policy reasons.
- **QDCOUNT**(Question count): 16-bit field that defines the number of entries in the question section.
- **ANCOUNT**(Answer count): 16-bit field that defines the number of resource records in the answer section.
- **NSCOUNT**(Authority count): 16-bit field that defines the number of name server resource records in the authority section.
- **ARCOUNT**(Additional count): 16-bit field that defines the number of resource records in the additional records section.



- **QUERY TYPE**: 16-bit field used to specify the type of the query
 - A(1) – Host address
 - MX(15) – Mail exchanger for the domain
 - ...
- **QUERY CLASS**: 16-bit field used to specify the class of the query
 - IN(1) – Internet system
 - ...

17

Resource Record Format (1)





■ **SOA**

- Start Of Authority--identifies the domain or zone and sets a number of parameters

■ **NS**

- Maps a domain name to the name of a computer that is authoritative for the domain

■ **A**

- Maps the name of a system to its address. If a system (e.g., a router) has several addresses, then there will be a separate record for each.

■ **AAAA**

- Maps the name of a system to its IPv6 address. If a system (e.g., a router) has several addresses, then there will be a separate record for each.

■ **CNAME**

- Maps an alias name to the true, canonical name

■ **MX**

- Mail Exchanger. Identifies the systems that relay mail into the organization

❖ **TXT**

- Provides a way to add text comments to the database. For example, a txt record could map abc.com to the company's name, address, and telephone number

❖ **WKS**

- Well Known Services. Can list the application services available at the host. Used sparingly, if at all

❖ **HINFO**

- Host Information, such as computer type and model. Rarely used

❖ **PTR**

- Maps an IP address to a system name. Used in address-to-name files.

■ Type=A

- Name= Domain name , Value= IP Address

ns.bupt.edu.cn 86400 IN A 202.112.10.37

■ Type=NS

- Name= Domain, eg. bupt.edu.cn
- value= Domain name of Authoritative Name Server

bupt.edu.cn 86400 IN NS ns.bupt.edu.cn

- Type=CNAME

- Name= canonical name
- Value= the name

dns.bupt.edu.cn 86400 IN CNAME ns.bupt.edu.cn

- Type=MX

- Name= Domain name
- Value= canonical name of mail server

bupt.edu.cn 86400 IN MX mail.bupt.edu.cn

■ Query type=A

```
命令提示符
C:\Users\DELL>nslookup www.bupt.edu.cn
服务器: UnKnown
Address: 10.3.9.44

非权威应答:
名称: vn46.bupt.edu.cn
Addresses: 2001:da8:215:4038::161
           10.3.9.161
Aliases: www.bupt.edu.cn

C:\Users\DELL>
```

■ Query type=MX

```
命令提示符
C:\Users\DELL>nslookup -qt=MX mail.bupt.edu.cn
服务器: UnKnown
Address: 10.3.9.44

非权威应答:
mail.bupt.edu.cn canonical name = ssl.exmail.qq.com

exmail.qq.com
primary name server = ns-tel1.qq.com
responsible mail addr = webmaster.qq.com
serial = 1277374542
refresh = 300 (5 mins)
retry = 600 (10 mins)
expire = 86400 (1 day)
default TTL = 300 (5 mins)
```



- **Wireshark: 网络协议分析器 (packet sniffer)**
- 能够在网络上抓消息包，显示消息包字段和其含义
- 用于网络故障排除、分析、软件和通信协议开发以及教育

■ 官网网址

- <http://www.wireshark.org/>

■ 下载网址

- <http://www.wireshark.org/download.html>

在Windows环境中Wireshark软件



以管理员身份运行，*Select Interface*, set *Capture Filter*, then press *Start*

The Wireshark Network Analyzer [Wireshark 1.12.5 (v1.12.5-0-g5819e5b from master-1.12)]

File Edit View Go Capture Analyze Statistics Telephony Tools Internals Help

Filter: Expression... Clear Apply Save

WIRESHARK

The Version

Capture

Interface List

Live list of the capture interfaces (counts incoming packets)

Start

Choose one or more interfaces to capture

- 本地连接
- VirtualBox Host-Only Network
- 无线网络连接

Capture Options

Start a capture with detailed options

Capture Hel

Wireshark: Capture Options

Capture

Capture	Interface	Link-layer header	Prom. Mode	Snaptlen [B]	Buffer [MiB]	
☒	无线网络连接 fe80-e4d3-e568-f54c-3c90 192.168.0.104	Ethernet	enabled	262144	2	udp port 53

☐ Capture on all interfaces

☒ Use promiscuous mode on all interfaces

Capture Filter: udp port 53

Capture Files

File: Browse...

☐ Use multiple files ☒ Use pcap-ng format

☒ Next file every 1 megabyte(s)

☐ Next file every 1 minute(s)

☐ Ring buffer with 2 files

在Windows环境中Wireshark软件



以管理员身份运行， *Select Interface*, set *Capture Filter*, then press *Start*

文件(F) 编辑(E) 视图(V) 跳转(G) 捕获(C) 分析(A) 统计(S) 电话(Y) 无线(W) 工具(T) 帮助(H)



应用显示过滤器 ... <Ctrl-/>

No.	Source	Destination	Protocol	Length	Info
49496600	192.168.1.4	120.92.210.67	TCP	54	52674 → 80 [ACK] Seq=1 Ack=1 Win=131840 Len=0
66739200	192.168.1.4	120.92.210.67	HTTP	106	GET / HTTP/1.0
92299700	120.92.210.67	192.168.1.4	TCP	54	80 → 52674 [ACK] Seq=1 Ack=53 Win=29184 Len=0
14116800	120.92.210.67	192.168.1.4	HTTP	213	HTTP/1.1 200 OK
14117600	120.92.210.67	192.168.1.4	TCP	54	80 → 52674 [FIN, ACK] Seq=160 Ack=53 Win=29184 Len=0
14180900	192.168.1.4	120.92.210.67	TCP	54	52674 → 80 [ACK] Seq=53 Ack=161 Win=131584 Len=0
14261400	192.168.1.4	120.92.210.67	TCP	54	52674 → 80 [FIN, ACK] Seq=53 Ack=161 Win=131584 Len=0
14522000	192.168.1.4	110.43.81.126	TCP	66	52675 → 80 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 WS=256 SACK_P...
25231800	120.92.210.67	192.168.1.4	TCP	54	80 → 52674 [ACK] Seq=161 Ack=54 Win=29184 Len=0
34047900	192.168.1.4	192.168.1.1	DNS	73	Standard query 0x2233 A www.baidu.com
39305600	192.168.1.1	192.168.1.4	DNS	132	Standard query response 0x2233 A www.baidu.com CNAME www.a.sh...
23056500	192.168.1.4	110.43.81.126	TCP	66	[TCP Retransmission] 52675 → 80 [SYN] Seq=0 Win=64240 Len=0 M...
25348400	192.168.1.4	110.43.81.126	TCP	66	[TCP Retransmission] 52675 → 80 [SYN] Seq=0 Win=64240 Len=0 M...
27901300	192.168.1.4	110.43.81.126	TCP	66	52677 → 80 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 WS=256 SACK_P...
47656000	110.43.81.126	192.168.1.4	TCP	66	80 → 52677 [ACK] Seq=161 Ack=54 Win=29184 Len=0

> Frame 215: Packet, 54 bytes on wire (432 bits), 54 bytes capture
> Ethernet II, Src: Intel_87:4a:d3 (f8:94:c2:87:4a:d3), Dst: Fiber
> Internet Protocol Version 4, Src: 192.168.1.4, Dst: 120.92.210.6
> Transmission Control Protocol, Src Port: 52674, Dst Port: 80, Se

0000 08 74 58 6f df c0 f8 94 c2 87 4a d3 08 00 45 00 ·tXo····
0010 00 28 f1 e3 40 00 40 06 3c a0 c0 a8 01 04 78 5c ·(·@·@·<
0020 d2 43 cd c2 00 50 fa 3f cc 05 f0 63 6a c6 50 10 ·C··P·?·
0030 02 03 b2 02 00 00 ······

■ DNS

- 启动Wireshark
- 可以在cmd命令行环境中运行 *nslookup* 命令

DNS Request



12	W2K3-SERVER	[202.106.0.20]	DNS: C ID=34422 OP=QUERY NAME=ftp.bupt.edu.cn	75	0:00:17.779	4.061.449	2010-08-06
13	[202.106.0.20]	W2K3-SERVER	DNS: R ID=34422 OP=QUERY STAT=OK NAME=ftp.bu	91	0:00:17.791	0.011.661	2010-08-06
14	W2K3-SERVER	ftp.bupt.edu.cn	ICMP: Echo	1514	0:00:17.812	0.021.578	2010-08-06

```

IP:
UDP: ----- UDP Header -----
UDP:
UDP: Source port      = 3017
UDP: Destination port = 53 (Domain)
UDP: Length           = 41
UDP: Checksum         = 3B76 (correct)
UDP: [33 byte(s) of data]
UDP:
DNS: ----- Internet Domain Name Service header -----
DNS:
DNS: ID = 34422
DNS: Flags = 01
DNS: 0... .. = Command
DNS: .000 0... = Query
DNS: .... ..0. = Not truncated
DNS: .... ..1 = Recursion desired
DNS: Flags = 0X
DNS: ...0 .... = Non Verified data NOT acceptable
DNS: Question count = 1, Answer count = 0
DNS: Authority count = 0, Additional record count = 0
DNS:
DNS: ZONE Section
DNS:   Name = ftp.bupt.edu.cn
DNS:   Type = Host address (A,1)
DNS:   Class = Internet (IN,1)
DNS:
  
```

```

00000000: 00 23 cd 82 e0 f6 00 0f 1f 52 ef f6 08 00 45 00 .#翊園...R稀..E.
00000010: 00 3d 01 82 00 00 80 11 ad ff c0 a8 00 08 ca 6a .=?..?括..黄
00000020: 00 14 0b c9 00 35 00 29 3b 76 86 76 01 00 00 01 ...?5.)v 啾....
00000030: 00 00 00 00 00 00 03 66 74 70 04 62 75 70 74 03 .....ftp.bupt.
00000040: 65 64 75 02 63 6e 00 00 01 00 01                edu.cn...
  
```

DNS Response



13	[202.106.0.20]	W2K3-SERVER	DNS: R ID=34422 OP=QUERY STAT=OK NAME=ftp.bupt.edu.cn	91	0:00:17.791	0.011.661	2010-08-06
14	W2K3-SERVER	ftp.bupt.edu.cn	ICMP: Echo	1514	0:00:17.812	0.021.578	2010-08-06
15	W2K3-SERVER	ftp.bupt.edu.cn	IP: Continuation of frame 14; 348 Bytes of data	362	0:00:17.812	0.000.085	2010-08-06

```

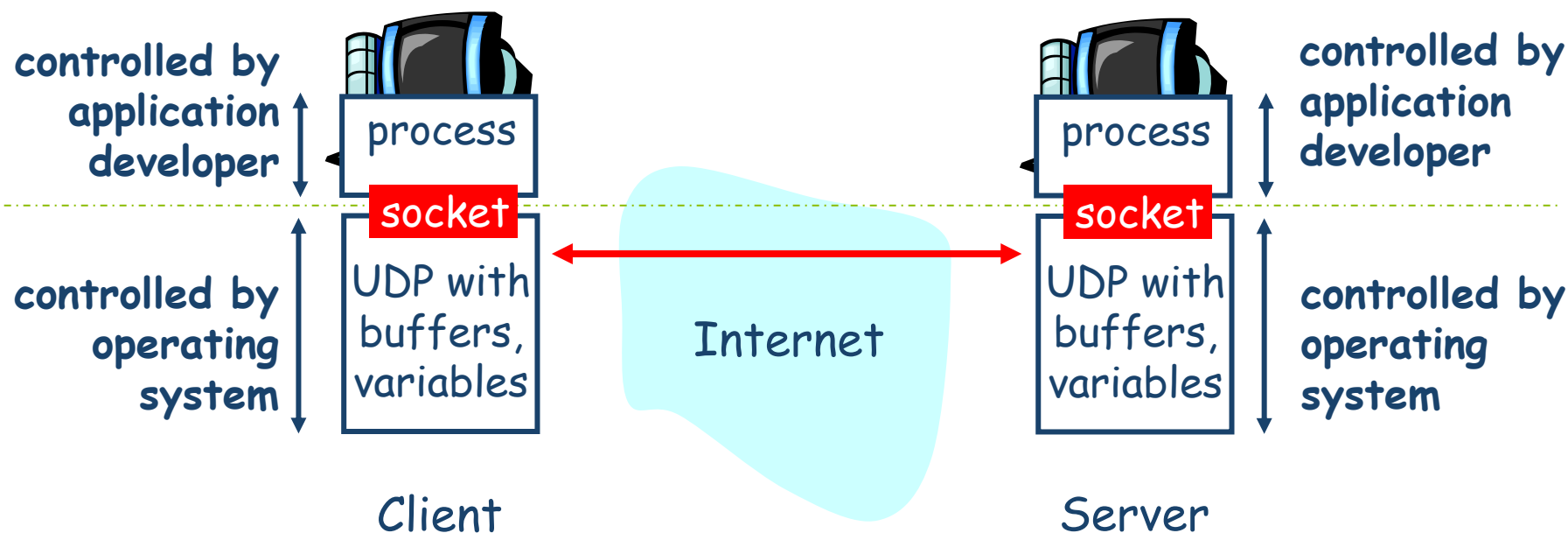
UDP:
DNS: ----- Internet Domain Name Service header -----
DNS:
DNS: ID = 34422
DNS: Flags = 81
DNS: 1... = Response
DNS: ...0... = Not authoritative answer
DNS: ...000 0... = Query
DNS: ...0... = Not truncated
DNS: Flags = 8X
DNS: ...0... = Data NOT verified
DNS: 1... = Recursion available
DNS: Response code = OK (0)
DNS: ...0... = Unicast packet
DNS: Question count = 1, Answer count = 1
DNS: Authority count = 0, Additional record count = 0
DNS:
DNS: ZONE Section
DNS: Name = ftp.bupt.edu.cn
DNS: Type = Host address (A,1)
DNS: Class = Internet (IN,1)
DNS:
DNS: Answer section:
DNS: Name = ftp.bupt.edu.cn
DNS: Type = Host address (A,1)
DNS: Class = Internet (IN,1)
DNS: Time-to-live = 6426 (seconds)
DNS: Length = 4
DNS: Address = [211.68.71.80]
DNS:
  
```

```

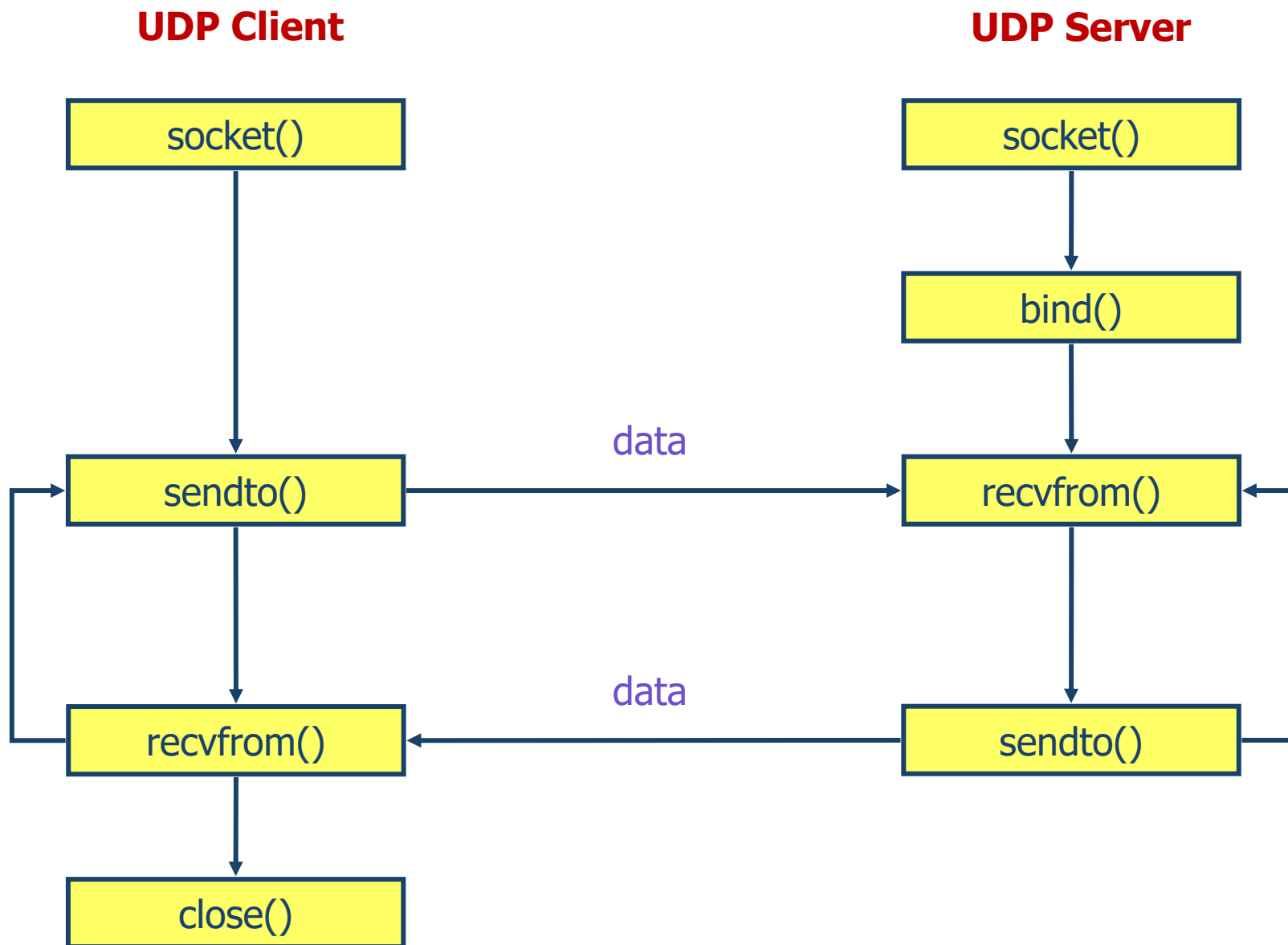
00000000: 00 0f 1f 52 ef f6 00 23 cd 82 e0 f6 08 00 45 00 ...R梯.#蛸園..E
00000010: 00 4d 29 cc 00 00 3a 11 cb a5 ca 6a 00 14 c0 a8 (M)?...衰資...括
00000020: 00 08 00 35 0b c9 00 39 f8 e0 86 76 81 80 00 01 ...5.29 嗽亏..
00000030: 00 01 00 00 00 00 03 66 74 70 04 62 75 70 74 03 .....ftp.bupt.
00000040: 65 64 75 02 63 6e 00 00 01 00 01 c0 0c 00 01 00 edu.cn.....?...
00000050: 01 00 00 19 1a 00 04 d3 44 47 50 .....視GP
  
```

- **Socket**套接口：操作系统**I/O**系统的扩展，用于实现进程间及机器间的通信
- 一种接口（“门”），应用程序进程可通过其向另一（远程或本地）应用程序进程**发送和接收**消息

- **Socket套接口**：应用程序进程与端到端传输协议（**UDP或TCP**）之间的接口（即“门”）
- **UDP服务**：实现进程间数据包的不可靠传输



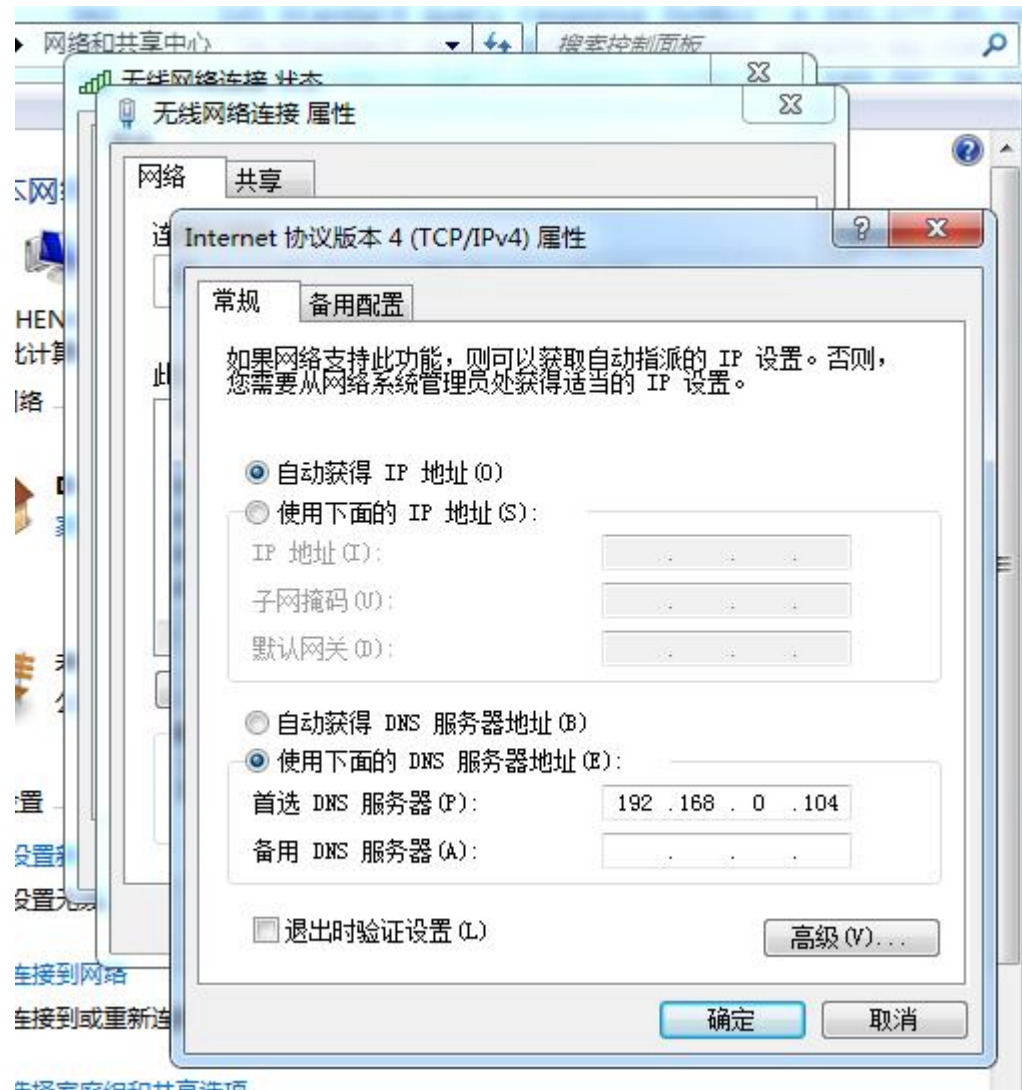
基于UDP的Socket API概述



- 1. Get the IP address of your DNS server (eg. *202.106.0.20*) and your IP address (eg. *192.168.0.104*)

c:> ipconfig /all

- 2. configure your DNS server as your IP address (eg. *192.168.0.104* or *127.0.0.1*)



运行Client-Server Socket通信程序(2)



命令提示符

D:\python_work>cd 10_Socket_Client-Server

D:\python_work\10_Socket_Client-Server>dir

驱动器 D 中的卷是 新加卷
卷的序列号是 5028-A885

D:\python_work\10_Socket_Client-Server 的目录

2026/06/28	15:30	<DIR>	.
2026/06/28	15:30	<DIR>	..
2026/06/28	14:29		902 client.py
2026/06/28	15:33	<DIR>	CSDN
2026/06/28	12:42	<DIR>	DeepSeek_V4
2026/06/28	14:28		1,449 server.py
		2 个文件	2,351 字节
		4 个目录	80,153,915,392 可用字节

D:\python_work\10_Socket_Client-Server>python client.py

已成功连接到服务器! 输入'exit' 退出连接

请输入要发送的消息: Hello, server 2026!

服务端回复: 服务端已收到: Hello, server 2026!

请输入要发送的消息: Hello, July 2026!

服务端回复: 服务端已收到: Hello, July 2026!

请输入要发送的消息: exit

已断开与服务端的连接

D:\python_work\10_Socket_Client-Server>

搜索



明天有暴雨

15:48
2026/6/28

1

运行Client-Server Socket通信程序(2)



命令提示符 - server.py

```
2022/08/05 18:37 1,520,422 python 中如何Debug_测试小白球的博客-CSDN博客_python代码debug.png
2025/12/08 10:57 11,505 python学习心得.docx
4 个文件 3,407,417 字节
12 个目录 80,153,915,392 可用字节
```

D:\python_work>cd 10_Socket_Client-Server

D:\python_work\10_Socket_Client-Server>dir

驱动器 D 中的卷是 新加卷
卷的序列号是 5028-A885

D:\python_work\10_Socket_Client-Server 的目录

```
2026/06/28 15:30 <DIR> .
2026/06/28 15:30 <DIR> ..
2026/06/28 14:29 902 client.py
2026/06/28 15:33 <DIR> CSDN
2026/06/28 12:42 <DIR> DeepSeek_V4
2026/06/28 14:28 1,449 server.py
2 个文件 2,351 字节
4 个目录 80,153,915,392 可用字节
```

D:\python_work\10_Socket_Client-Server>server.py

服务端已启动, 正在监听端口 80 ...

客户端已连接: ('127.0.0.1', 51591)

收到客户端消息: Hello,server 2026!

收到客户端消息: Hello, July 2026!

客户端已断开连接



运行DNS Relay client-server程序(2)



命令提示符

C:\Users\Laptop>d:

D:\>cd python_work

D:\python_work>cd DNS_relay

D:\python_work\DNS_relay>dir

驱动器 D 中的卷是 新加卷
卷的序列号是 5028-A885

D:\python_work\DNS_relay 的目录

```
2026/06/30 14:58 <DIR> .
2026/06/30 14:58 <DIR> ..
2026/06/30 14:57 <DIR> Client-Server
2026/06/30 15:36      1,206 client.py
2026/06/28 15:33 <DIR> CSDN
2026/06/28 12:42 <DIR> DeepSeek_V4
2026/06/30 15:16      1,705 dnsrelay.py
2021/07/04 14:52      59 dnsrelay.txt
2026/06/30 14:55      1,186 relay.py
               4 个文件      4,156 字节
               5 个目录 80,175,501,312 可用字节
```

D:\python_work\DNS_relay>python client.py

客户端: 向DNS中继发送查询 www.baidu.com 请求

客户端收到中继返回的DNS原始报文(十六进制):

2233818000010003000000000037777705626169647503636f6d0000010001c00c0005000100000150000f0377777701610673686966656ec016c02b000100010000003c00046ef24515c02b000100100000003c00046ef24639

D:\python_work\DNS_relay>python client.py

客户端: 向DNS中继发送查询 www.baidu.com 请求

客户端收到中继返回的DNS原始报文(十六进制):

2233818000010003000000000037777705626169647503636f6d0000010001c00c0005000100000131000f0377777701610673686966656ec016c02b000100010000001d00046ef24639c02b000100100000001d00046ef24515

D:\python_work\DNS_relay>dig @127.0.0.1 -p 5353 www.baidu.com

'dig' 不是内部或外部命令，也不是可运行的程序
或批处理文件。

D:\python_work\DNS_relay>



运行DNS Relay client-server程序(2)



命令提示符 - python dnsrelay.py

4 个文件 3,407,417 字节
13 个目录 80,175,501,312 可用字节

D:\python_work>cd DNS_relay

D:\python_work\DNS_relay>dir

驱动器 D 中的卷是 新加卷
卷的序列号是 5028-A885

D:\python_work\DNS_relay 的目录

```
2026/06/30 14:58 <DIR> .
2026/06/30 14:58 <DIR> ..
2026/06/30 14:57 <DIR> Client-Server
2026/06/30 15:36      1,206 client.py
2026/06/28 15:33 <DIR> CSDN
2026/06/28 12:42 <DIR> DeepSeek_V4
2026/06/30 15:16      1,705 dnsrelay.py
2021/07/04 14:52         59 dnsrelay.txt
2026/06/30 14:55      1,186 relay.py
4 个文件 4,156 字节
5 个目录 80,175,501,312 可用字节
```

D:\python_work\DNS_relay>python dnsrelay.py

DNS Relay running on 127.0.0.1:53, upstream DNS: 192.168.1.1:53

Receive DNS query from ('127.0.0.1', 51446)

Forward DNS response to ('127.0.0.1', 51446)

Receive DNS query from ('127.0.0.1', 57605)

Forward DNS response to ('127.0.0.1', 57605)

-

英 : 语音 窗口 任务 设置 帮助



搜索



26°C 多云



16:14

2026/6/30

■ Report

- 每个班的学委收齐课程设计报告并发送email到邮箱 ly@bupt.edu.cn